

Networking Basics

What is an IP Address

An IP address is a unique numerical label assigned to every device connected to a computer network. It allows devices to locate and communicate with each other. IPv4 addresses are written as four numbers separated by dots, such as 192.168.1.1, while IPv6 addresses are longer and written in hexadecimal groups.

What is a Router

A router is a networking device that forwards data packets between computer networks. It connects a local network, such as a home or office, to the wider internet, and directs incoming and outgoing traffic to the correct destination device based on IP addresses.

What is Bandwidth

Bandwidth refers to the maximum rate of data transfer across a network path, usually measured in megabits per second (Mbps) or gigabits per second (Gbps). Higher bandwidth allows more data to be transmitted in a given amount of time, which improves speed for tasks like streaming and large file downloads.

What is Latency

Latency is the time delay between sending a request over a network and receiving a response. It is typically measured in milliseconds (ms). Low latency is important for real-time applications such as video calls and online gaming, where delays are noticeable to users.

What is a Firewall

A firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predetermined security rules. It acts as a barrier between a trusted internal network and untrusted external networks, such as the internet.